

Nowhere-Surjective Linearization of the Global Jacobian and the Failure of Locally Lipschitz Sections

Partial-result packet for arXiv:1611.02223

11 August 2026

Status: candidate substantial partial result, likely valid; human review requested. For every $n \geq 2$, every $1 \leq p < \infty$, and every base map u , the derivative of the global Jacobian $J : \dot{W}^{1,np} \rightarrow \mathcal{H}^p$ is not onto. Consequently no right inverse is Gâteaux differentiable anywhere, and no locally Lipschitz right inverse exists, even locally over a nonempty open target set. The argument includes the exponent $p = 2$ omitted by the known general Banach-geometry obstruction. Bare surjectivity and merely continuous right inverses remain open.

1 Source problem

The source is Sauli Lindberg, *On the Hardy Space Theory of Compensated Compactness Quantities*, arXiv:1611.02223, published in 2017 [1]. Its Conjecture 7.1, attributed to Iwaniec, states:

For every $n \geq 2$ and every $p \in [1, \infty)$, the Jacobian operator

$$J : \dot{W}^{1,np}(\mathbb{R}^n, \mathbb{R}^n) \longrightarrow \mathcal{H}^p(\mathbb{R}^n)$$

has a continuous right inverse.

Here $\mathcal{H}^1$ is the real Hardy space and $\mathcal{H}^p = L^p$ with equivalent norms when $p > 1$. The source emphasizes that the homogeneous scaling removes its inhomogeneous non-surjectivity argument.

Later work proves that surjectivity is equivalent to a bounded selection and to the natural a priori estimate, but the problem remains open [3]. That work also rules out Fréchet-differentiable right inverses for $p \neq 2$ by Banach-space geometry. The result below is specific to the Jacobian: it covers every p , strengthens Fréchet to Gâteaux differentiability, and yields a locally Lipschitz no-go theorem.

2 Proof intuition

At a map u , the Jacobian linearizes to

$$DJ(u)v = \text{cof}(Du) : Dv.$$

The coefficient $\text{cof}(Du)$ has a finite global Lebesgue norm. Translate one fixed smooth dual test bump into pairwise disjoint regions. Its dual norm is unchanged, while the cofactor mass in those regions tends to zero. Hence the adjoint of $DJ(u)$ sends unit dual vectors to zero in norm. The adjoint of a surjective bounded linear map is bounded below, so $DJ(u)$ cannot be onto.

The basic idea of Proposition 6.3 is that the integration by parts creates cancellations of partial derivatives of v but such cancellations are not present in integrals $\int_{\mathbb{R}^3} \varphi \mathcal{L}(\nabla^2(u, v))$, where $\varphi \in C_c^\infty(\mathbb{R}^3)$.

7. OPEN PROBLEMS RELATED TO QUESTION 1.1

In this section we discuss open questions on the surjectivity of the Jacobian operator in Hardy spaces. The core of all the non-surjectivity results of this article is the incompatibility of the scaling properties of inhomogeneous Sobolev spaces and the target space $\mathcal{H}^1(\mathbb{R}^n)$, and this particular obstacle to surjectivity is not present when homogeneous Sobolev spaces are used as domains of definition of the operators or the surjectivity question is treated in bounded domains.

7.1. Homogeneous Sobolev spaces. Coifman, Lions, Meyer, and Semmes deduced Theorem 2.2 by proving the norm equivalence

$$(7.1) \quad \|b\|_{\text{BMO}} \approx \sup \left\{ \int_{\mathbb{R}^n} bJu : u \in W^{1,n}(\mathbb{R}^n, \mathbb{R}^n), \int_{\mathbb{R}^n} |Du|^n \leq 1 \right\}$$

and using direction (i) $\Rightarrow$ (iii) of Lemma 3.1. In view of (7.1) it is natural to ask whether J maps the inhomogeneous Sobolev space $\dot{W}^{1,n}(\mathbb{R}^n, \mathbb{R}^n)$ onto $\mathcal{H}^1(\mathbb{R}^n)$. Recall that when $1 < p < \infty$, the Hardy space $\mathcal{H}^p(\mathbb{R}^n)$ coincides with $L^p(\mathbb{R}^n)$. The following conjecture was posed by T. Iwaniec in 1997.

Conjecture 7.1 ([20]). *For every $n \geq 2$ and every $p \in [1, \infty[$ the Jacobian operator $J: \dot{W}^{1,np}(\mathbb{R}^n, \mathbb{R}^n) \rightarrow \mathcal{H}^p(\mathbb{R}^n)$ has a continuous right inverse.*

In [29] the author made progress on Conjecture 7.1 in the case $n = 2, p = 1$ via calculus of variations and Banach space geometry. Recall the norm $\|\cdot\|_{\text{BMO}_\infty}$

Figure 1: Section 7.1 and Conjecture 7.1 in the source PDF, page 25.

If a locally Lipschitz section existed, Phelps's infinite-dimensional Rademacher theorem would make it Gâteaux differentiable somewhere: the target of the section is a reflexive homogeneous Sobolev space and therefore has the Radon–Nikodým property. Differentiating $J \circ G = \text{Id}$ would force one of the impossible linearizations to be onto.

3 Statement

For $1 \leq p < \infty$, set

$$X_p = \dot{W}^{1,np}(\mathbb{R}^n, \mathbb{R}^n), \quad Y_p = \begin{cases} \mathcal{H}^1(\mathbb{R}^n), & p = 1, \\ L^p(\mathbb{R}^n), & p > 1. \end{cases}$$

The norm on X_p is $\|u\|_{X_p} = \|Du\|_{L^{np}}$, modulo constants.

Theorem 1 (Nowhere-surjective linearization). *Let $n \geq 2$ and $1 \leq p < \infty$. For every $u \in X_p$, the Fréchet derivative*

$$DJ(u) : X_p \longrightarrow Y_p$$

is not surjective.

Corollary 2 (No regular local selection). *Under the hypotheses of Theorem 1:*

1. *every set-theoretic right inverse $G : Y_p \rightarrow X_p$, if one exists, is Gâteaux-differentiable nowhere;*
2. *there do not exist a nonempty open set $U \subset Y_p$ and a locally Lipschitz map $G : U \rightarrow X_p$ such that $J(G(f)) = f$ for every $f \in U$.*

4 Proof

4.1 The derivative

The determinant polarization and Hölder's inequality show, for $p > 1$, that $J : X_p \rightarrow L^p$ is a continuous n -homogeneous polynomial. At $p = 1$, the multilinear compensated-compactness estimate of Coifman–Lions–Meyer–Semmes gives the corresponding continuity into $\mathcal{H}^1$ [2, 1]. Thus in every case J is Fréchet differentiable and

$$A_u v := DJ(u)v = \text{cof}(Du) : Dv. \tag{1}$$

If

$$q = \frac{np}{n-1}, \quad s = (np)' = \frac{np}{np-1}, \tag{2}$$

then

$$C_u := \text{cof}(Du) \in L^q(\mathbb{R}^n; \mathbb{R}^{n \times n}).$$

4.2 Localized dual bumps

We prove Theorem 1. First suppose $p > 1$. Choose a nonzero $\psi \in C_c^\infty(\mathbb{R}^n)$ normalized by $\|\psi\|_{L^{p'}} = 1$. Select translations a_k so that

$$E_k = a_k + \text{supp } \psi$$

are pairwise disjoint, and set $\psi_k(x) = \psi(x - a_k)$. The dual vectors $\psi_k \in Y_p^* = L^{p'}$ all have norm one. For $v \in X_p$, equations (1)–(2) and Hölder's inequality give

$$|\langle A_u v, \psi_k \rangle| \leq \|\psi_k C_u\|_{L^s} \|Dv\|_{L^{np}}, \quad (3)$$

$$\|\psi_k C_u\|_{L^s} \leq \|\psi_k\|_{L^{p'}} \|1_{E_k} C_u\|_{L^q} = \|1_{E_k} C_u\|_{L^q}, \quad (4)$$

because $1/s = 1/p' + 1/q$. Since the E_k are disjoint and $C_u \in L^q$,

$$\|1_{E_k} C_u\|_{L^q} \longrightarrow 0. \quad (5)$$

It follows that

$$\|A_u^* \psi_k\|_{X_p^*} \longrightarrow 0 \quad \text{while} \quad \|\psi_k\|_{Y_p^*} = 1.$$

For $p = 1$, choose a nonconstant $\psi \in C_c^\infty(\mathbb{R}^n)$ normalized by $\|\psi\|_{\text{BMO}} = 1$, where BMO is understood modulo constants, and use the same disjoint translations. Translation invariance gives $\|\psi_k\|_{(\mathcal{H}^1)^*} = 1$. Here $q = s = n/(n-1)$, and the compact bump has finite L^∞ norm, so (3) becomes

$$\|A_u^* \psi_k\|_{X_1^*} \leq \|\psi\|_{L^\infty} \|1_{E_k} C_u\|_{L^{n/(n-1)}} \longrightarrow 0.$$

The pairing is the ordinary integral because the test functions are smooth and compactly supported.

For any bounded surjection $A : X \rightarrow Y$, the open mapping theorem implies that $A^* : Y^* \rightarrow X^*$ is bounded below. The normalized sequences above contradict that necessary condition. Therefore $A_u = DJ(u)$ is not onto for any u , proving Theorem 1.

4.3 Right inverses

Suppose a right inverse G is Gâteaux differentiable at f . Since J is Fréchet differentiable, the chain rule applied to $J \circ G = \text{Id}$ gives

$$DJ(G(f)) DG(f) = I_{Y_p}.$$

Thus $DJ(G(f))$ is onto, contrary to Theorem 1. This proves the first part of Corollary 2.

For the second part, Y_p is separable. Moreover, the gradient identifies X_p isometrically with a closed subspace of L^{np} ; since $np > 1$, X_p is reflexive and hence has the Radon–Nikodým property. Phelps's theorem therefore says that a locally Lipschitz map from a nonempty open subset of Y_p into X_p is Gâteaux differentiable outside a Gaussian-null set, in particular at some point [4]. A locally Lipschitz local section would contradict the first part. This completes the proof.

5 What this settles and what remains open

The theorem closes the $p = 2$ gap in the known *linearization* obstruction and rules out a broad, natural regularity class of selections. The proof is local and requires no assumption that the full Jacobian is onto.

It does not settle the source conjecture. A merely continuous right inverse need not be Gâteaux differentiable anywhere, and the natural scaling of an n -homogeneous equation suggests only $1/n$ -Hölder behavior at the origin. Nor does non-surjectivity of every derivative imply non-surjectivity of a nonlinear map in infinite dimensions. Thus bare surjectivity, continuous selection, the source's sharper energy-minimizer conjecture, and the bounded- domain variants all remain open.

Verification notes. The proof has no computational dependency. The main checkpoints are the Hardy-endpoint derivative estimate, the exponent identity $1/(np)' = 1/p' + (n-1)/(np)$, the use of pairwise disjoint translated supports, and the RNP hypothesis in Phelps's theorem.

Novelty check. On 11 August 2026 we searched the run’s four lightweight indexes and the exact source/older-paper titles, together with combinations of “linearized Jacobian not surjective”, “Jacobian submersion nowhere $p = 2$ ”, “locally Lipschitz right inverse Jacobian”, and “Phelps Jacobian right inverse”. The closest result found is the general Banach-geometry obstruction in [3]: no submersion for $p < 2$, no regular submersion for $p > 2$, and no Fréchet-differentiable right inverse for $p \neq 2$. No source located the cofactor-localization proof, the all- p statement, or the locally Lipschitz corollary. Novelty confidence is moderate, not definitive.

Human-review recommendation. Review as a substantial partial result. In addition to the four checkpoints above, compare carefully with the published version of [3] to confirm that the Jacobian-specific $p = 2$ observation and Phelps upgrade have not been recorded elsewhere.

References

- [1] S. Lindberg, *On the Hardy Space Theory of Compensated Compactness Quantities*, Arch. Ration. Mech. Anal. **224** (2017), 709–742; arXiv:1611.02223.
- [2] R. Coifman, P.-L. Lions, Y. Meyer, and S. Semmes, *Compensated Compactness and Hardy Spaces*, J. Math. Pures Appl. **72** (1993), 247–286.
- [3] A. Guerra, L. Koch, and S. Lindberg, *Nonlinear Open Mapping Principles, with Applications to the Jacobian Equation and Other Scale-Invariant PDEs*, arXiv:2010.10497 (2020).
- [4] R. R. Phelps, *Gaussian Null Sets and Differentiability of Lipschitz Maps on Banach Spaces*, Pacific J. Math. **77** (1978), 523–531.